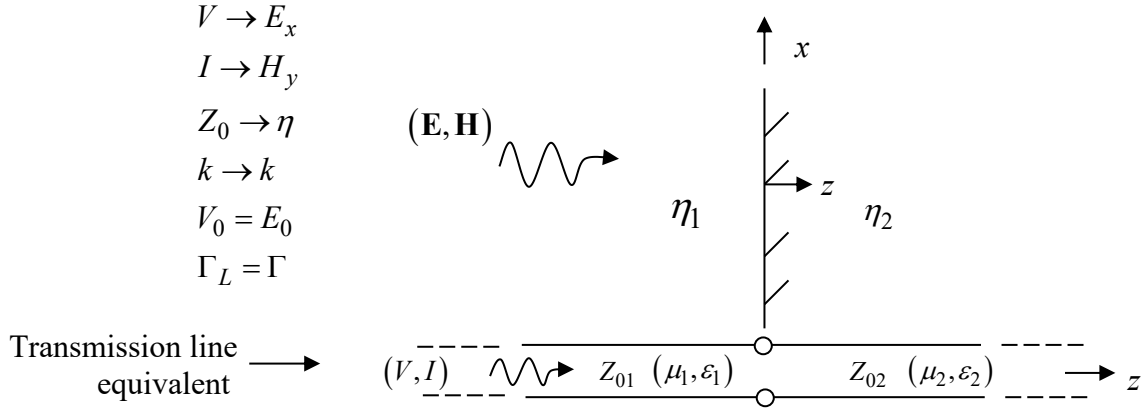


Notes 4

Transmission Line/Plane Wave Scatterer Equivalence Example

Now solve the problem of reflection from a dielectric interface with duals:



In wave scattering problems the coordinate system is placed at a dielectric boundary. The transmission line equivalent is the intersection of two transmission lines – not at the load, which in this case is infinity.

Transmission Line Equivalent

Field Dual

$$V_1 = V_0 e^{-jk_1 z} (1 + \Gamma_L e^{j2k_1 z}) \longleftrightarrow E_{x1} = E_0 e^{-jk_1 z} (1 + \Gamma e^{j2k_1 z})$$

$$I_1 = \frac{V_0}{Z_{01}} e^{-jk_1 z} (1 - \Gamma_L e^{j2k_1 z}) \longleftrightarrow H_{y1} = \frac{E_0}{\eta_1} e^{-jk_1 z} (1 - \Gamma e^{j2k_1 z})$$

$$\Gamma_L = \frac{Z_L - Z_{01}}{Z_L + Z_{01}} = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}} \longleftrightarrow \Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

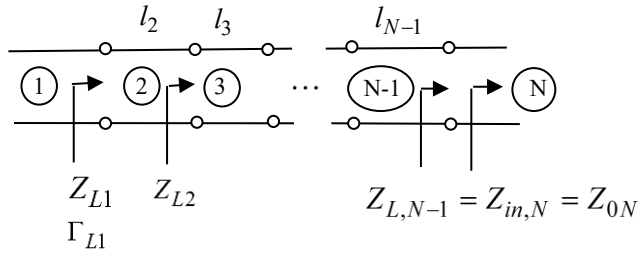
$$V_2 = T V_0 e^{-jk_2 z} \longleftrightarrow E_{x2} = T E_0 e^{-jk_2 z}$$

$$I_2 = T \frac{V_0}{Z_{02}} e^{-jk_2 z} \longleftrightarrow H_{y2} = \frac{T E_0}{\eta_2} e^{-jk_2 z}$$

$$T = \frac{2Z_{02}}{Z_{01} + Z_{02}} \longleftrightarrow T = \frac{2\eta_2}{\eta_1 + \eta_2} \quad \text{Transmission coefficient}$$

Redo this page with TL info on left hand side

- Multiple transmission lines



$$V_1 = V_{01} e^{-jk_1 z} (1 + \Gamma_{L1} e^{j2k_1 z})$$

$$V_n = V_{0n} e^{-jk_n(z-d_n)} (1 + \Gamma_{Ln} e^{j2k_n(z-d_n)}), \quad d_n = \sum_{m=1}^n l_m \quad \text{with } l_1 = 0$$

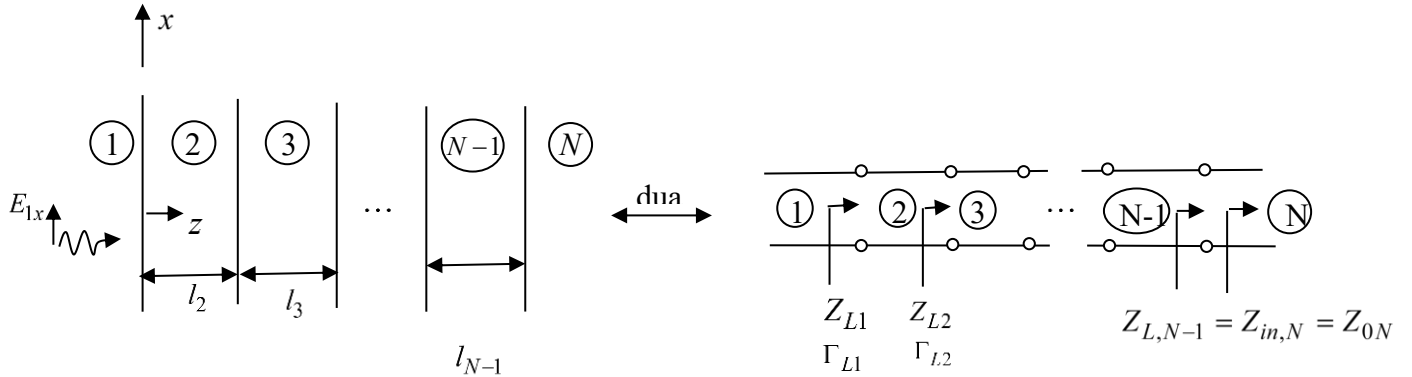
$$V_N = V_{0N} e^{-jk_N(z-d_{N-1})}$$

$$\Gamma_{Ln} = \frac{Z_{Ln} - Z_{on}}{Z_{Ln} + Z_{on}}$$

$$Z_{L,n} = Z_{o,n+1} \frac{Z_{L,n+1} + jZ_{o,n+1} \tan(k_{n+1}l_{n+1})}{Z_{o,n+1} + jZ_{L,n+1} \tan(k_{n+1}l_{n+1})}$$

$$Z_{L,N-1} = Z_{o,N}$$

- Multiple parallel plane layers



In region ①

$$E_{1x} = E_{01} e^{-jk_1 z} (1 + \Gamma_{L1} e^{j2k_1 z})$$

Γ_{L1} is found from:

$$\Gamma_{L1} = \frac{Z_{L1} - \eta_1}{Z_{L1} + \eta_1}$$

where

$$Z_{L1} = \eta_2 \frac{Z_{L2} + j\eta_2 \tan(k_2 l_2)}{\eta_2 + jZ_{L2} \tan(k_2 l_2)}$$

$$\vdots$$

$$V_1 = V_{01} e^{-jk_1 z} (1 + \Gamma_{L1} e^{j2k_1 z})$$

$$\Gamma_{L1} = \frac{Z_{L1} - Z_{o1}}{Z_{L1} + Z_{o1}}$$

$$Z_{L1} = Z_{02} \frac{Z_{L2} + jZ_{02} \tan(k_2 l_2)}{Z_{02} + jZ_{L2} \tan(k_2 l_2)}$$

$$\begin{aligned}
Z_{L,N-2} &= \eta_{N-1} \frac{Z_{L,N-1} + j\eta_{N-1} \tan(k_{N-1}l_{N-1})}{\eta_{N-1} + jZ_{L,N-1} \tan(k_{N-1}l_{N-1})} & \vdots \\
Z_{L,N-1} = Z_{in,N} = \eta_N & \leftarrow Z_{L,N-1} = Z_{in,N} = Z_{0,N} \\
& \text{characteristic impedance of last} \\
& \text{(infinite) section of line}
\end{aligned}$$

In region ②

$$E_{2x} = E_{02} e^{-jk_2(z-l_2)} \left(1 + \Gamma_{L2} e^{j2k_2(z-l_2)} \right)$$

Note: Γ_{L2} can be found from the impedances above and

E_{02} is found by requiring $E_{1x} = E_{2x}$ at $z = 0$

Therefore in a general region $n = 1, 2, 3, \dots, N-1$

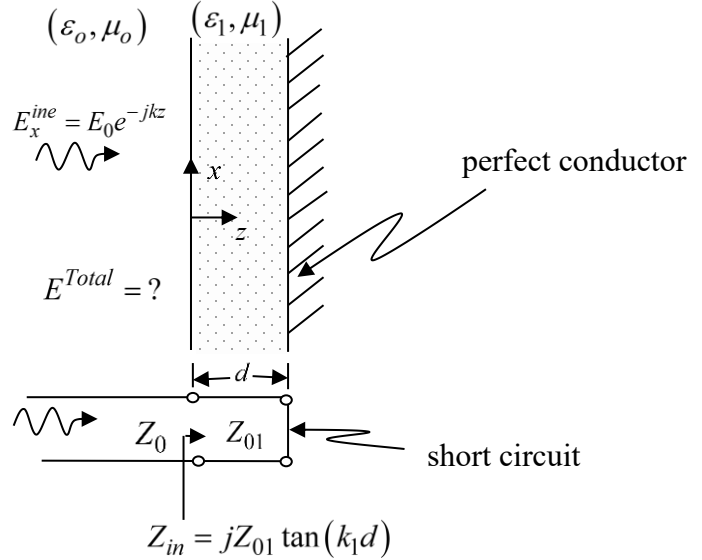
$$E_{nx} = E_{0n} e^{-jk_n(z-d_n)} \left(1 + \Gamma_{Ln} e^{j2k_n(z-d_n)} \right), \quad d_n = \sum_{m=1}^n l_m \quad \text{with } l_1 = 0$$

However in last region N, the wave is phase referenced to the last boundary

$$E_{Nx} = E_{0N} e^{-jk_N(z-d_{N-1})}$$

Where E_{0N} is found by requiring $(E_{1x} = E_{2x}|_{z=0}, \dots, E_{N-1,x} = E_{N,x}|_{z=d_{N-1}})$

Example: Find an expression for the total voltage in the Z_0 region in the transmission line. Then use duality to find the total electric field in the region in front of the grounded dielectric slab.

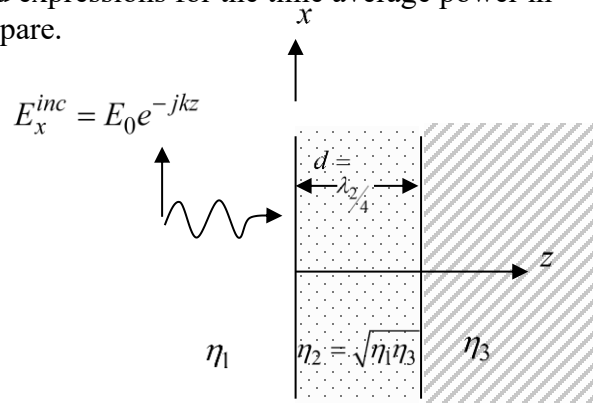


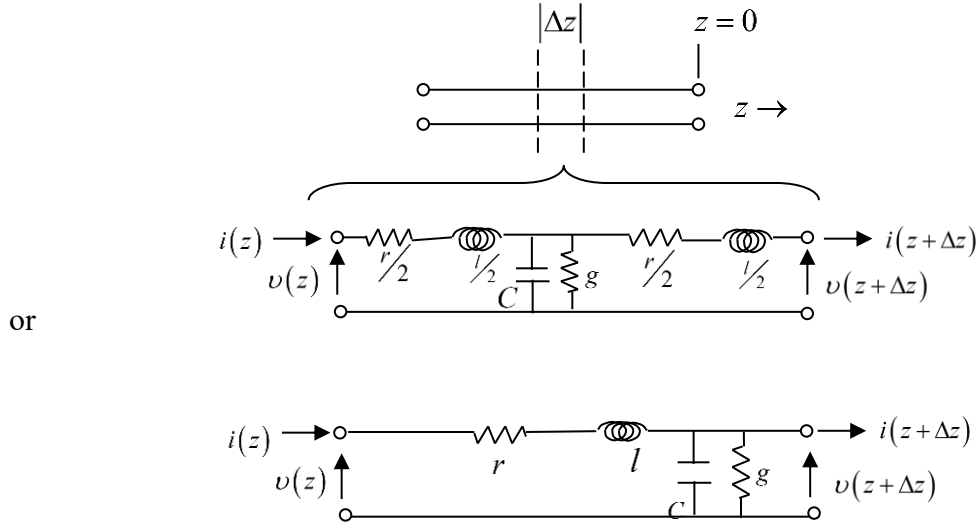
$$V = V_0 e^{-jkz} (1 + \Gamma_L e^{j2kz}) \rightarrow E_x = E_0 e^{-jkz} (1 + \Gamma e^{j2kz})$$

$$\Gamma_L = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} = \frac{jZ_{01} \tan(k_1 d) - Z_0}{jZ_{01} \tan(k_1 d) + Z_0} \rightarrow \Gamma = \frac{j\eta_1 \tan(k_1 d) - \eta_0}{j\eta_1 \tan(k_1 d) + \eta_0}$$

Homework: (a) Find an expression for the transmission line voltage in the short circuited transmission section $0 \leq z \leq d$ in the figure above. (b) Then use duality to find an expression for the electric field in the dielectric slab region $0 \leq z \leq d$. Define all variables in your answer in terms of given known quantities (e.g. E_0). Use the coordinate system defined in the figure above.

Homework: (a) Find expressions for the total E-fields in region 1 and 3 by using transmission line duals – solve the TL problem first then show the duals. Use the coordinate system shown and give your answer in terms of the given quantities. (b) Find expressions for the time average power in regions 1 and 3 and compare.





KVL: $v(z) - ri(z) - \ell \frac{di(z)}{dt} - v(z + \Delta z) = 0$

KCL: $i(z) - i(z + \Delta z) - c \frac{dv(z + \Delta z)}{dt} - gv(z + \Delta z) = 0$

rearrange and divide by Δz

$$\frac{v(z + \Delta z) - v(z)}{\Delta z} + \frac{r}{\Delta z} i(z) + \frac{\ell}{\Delta z} \frac{di(z)}{dt} = 0$$

$$\frac{i(z) - i(z + \Delta z)}{\Delta z} + \frac{g}{\Delta z} \cdot v(z + \Delta z) + \frac{c}{\Delta z} \frac{dv(z + \Delta z)}{dt} = 0$$

Take limit as $\Delta z \rightarrow 0$

$$\frac{dv(z)}{dz} + Ri(z) + L \frac{di(z)}{dt} = 0$$

$$\frac{di(z)}{dz} + Gv(z) + C \frac{dv(z)}{dt} = 0$$

where $\frac{r}{\Delta z} \rightarrow R(\Omega/m)$, $\frac{g}{\Delta z} \rightarrow G(\text{S}/m)$, $\frac{\ell}{\Delta z} \rightarrow L(H/m)$, $\frac{c}{\Delta z} \rightarrow C(F/m)$

Convert to frequency domain $v(z, t) \rightarrow V(z, \omega)$; $i(z, t) \rightarrow I(z, \omega)$, $\frac{d}{dt} \rightarrow j\omega$

$$\left. \begin{aligned} \frac{dV}{dz} &= -(R + j\omega L)I = -ZI \quad (1) \\ \frac{dI}{dz} &= -(G + j\omega C)V = -YV \quad (2) \end{aligned} \right\} \text{The transmission line equations}$$

Take derivative of (1) and substitute (2) into (1)

$$\frac{d^2 V}{dz^2} - \gamma^2 V = 0; \quad \gamma^2 = ZY \quad (3)$$

$$\text{where } \gamma^2 = ZY \text{ or } \gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

A solution to (3) is

$$V = V^+ e^{-\gamma z} + V^- e^{\gamma z} \quad (4)$$

Take derivative of (2) and substitute (1) into (2)

$$\frac{dI^2}{dz^2} - \gamma^2 I = 0; \quad \gamma^2 = ZY \quad (5)$$

The solution to (5) is

$$I = I^+ e^{-\gamma z} + I^- e^{\gamma z} \quad (6)$$

Substitute (4) into (1)

$$I = \frac{V^+}{\sqrt{\frac{R+j\omega L}{G+j\omega C}}} e^{-\gamma z} - \frac{V^-}{\sqrt{\frac{R+j\omega L}{G+j\omega C}}} e^{\gamma z} \quad (7)$$

Define:

$$Z_0 \doteq \sqrt{\frac{R+j\omega L}{G+j\omega C}} \text{ characteristic impedance}$$

Compare (6) and (7)

$$Z_0 = \frac{V^+}{I^+} = -\frac{V^-}{I^-}$$

In general we will not be concerned with loss

$$\Rightarrow R = 0, G = 0$$

$$\text{so } \gamma^2 = -\omega^2 LC \text{ and } Z_0 = \sqrt{L/C}$$

$$\equiv -k^2$$

The solution to (3) is then

$$V = V^+ e^{-jkz} + V^- e^{jkz}$$

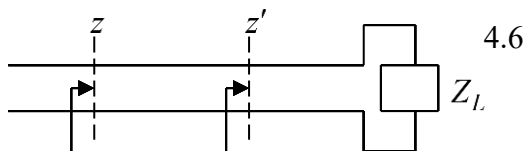
$$= V^+ e^{-jkz} \left(1 + \underbrace{\frac{V^-}{V^+} e^{j2kz}}_{\Gamma(z) \equiv \text{generalized reflection coefficient}} \right)$$

$\Gamma(z) \equiv \text{generalized reflection coefficient}$

$$V = V^+ e^{-jkz} (1 + \Gamma(z)) \quad (6)$$

and

$$I = \frac{V^+}{Z_0} e^{-jkz} (1 - \Gamma(z)) \quad (7)$$



$$\begin{aligned}
\Gamma(z) &= \Gamma_L e^{j2kz} \\
&= \Gamma_L e^{j2kz} e^{j2kz'} e^{-j2kz'} \\
&= \underbrace{\Gamma_L e^{j2kz'}}_{\Gamma(z')} e^{j2k(z-z')} \\
\Gamma(z) &= \Gamma(z') e^{j2k(z-z')} \quad (8)
\end{aligned}$$

Take ratio of (6) and (7)

$$Z(z) = Z_0 \left(\frac{1 + \Gamma(z)}{1 - \Gamma(z)} \right) \equiv \text{Line impedance} \quad (9)$$

or rearranging

$$\Gamma(z) = \frac{Z(z) - Z_0}{Z(z) + Z_0} \quad (10)$$

Example: At the load $Z(0) = Z_L$

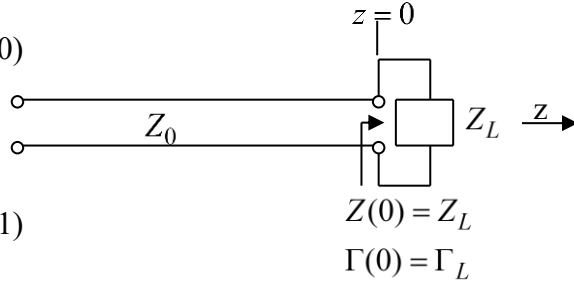
$$\Gamma(0) = \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (11)$$

Another way of writing (9) is

$$Z(z) = Z_0 \left(\frac{1 + \Gamma_L e^{j2kz}}{1 - \Gamma_L e^{j2kz}} \right) \quad (12)$$

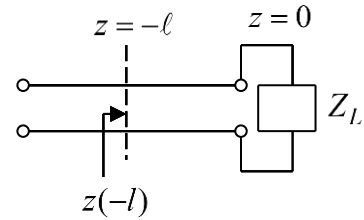
Substitute (11) into (12) and rearrange

$$Z(z) = Z_0 \left(\frac{Z_L - jZ_0 \tan(kz)}{Z_0 - jZ_L \tan(kz)} \right)$$



Z_L Or since z is always negative let $z = -\ell$

$$Z(-\ell) = Z_0 \left(\frac{Z_L + jZ_0 \tan(k\ell)}{Z_0 + jZ_L \tan(k\ell)} \right) \quad (13)$$

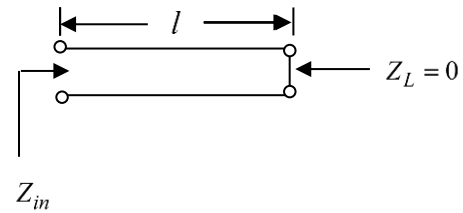


- Important special cases

Case 1: Short circuited line

$$Z(-l) = Z_0 \left(\frac{0 + jZ_0 \tan kl}{Z_0 + 0} \right)$$

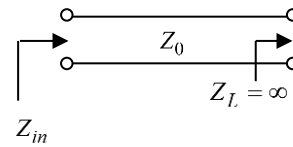
$$Z_{in} = jZ_0 \tan kl$$



Case 2: Open circuited line $\Rightarrow Z_L = \infty$

$$Z(-l) = Z_0 \frac{1 + j \frac{Z_0}{Z_L} \tan kl}{\frac{Z_0}{Z_L} + j \tan(kl)}$$

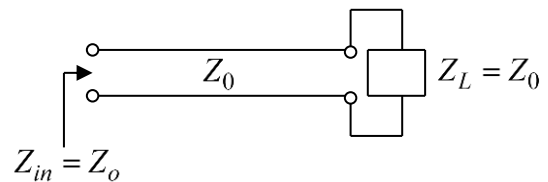
$$Z_{in} = -jZ_0 \cot(kl)$$



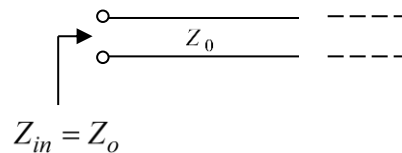
Case 3: Matched Line $\Rightarrow Z_L = Z_0$

$$Z(-l) = Z_0 \left(\frac{Z_0 + jZ_0 \tan(kl)}{Z_0 + jZ_0 \tan(kl)} \right)$$

$$Z_{in} = Z_0$$



Case 4: Infinite line



Voltage Standing wave ratio

$$VSWR = \frac{|V|_{\max}}{|V|_{\min}}$$

Recall

$$V = V_+ e^{-jkz} (1 + \Gamma(z))$$

so

$$|V| = |V_+| |1 + \Gamma(z)|$$

$$\text{But } \Gamma(z) = |\Gamma_L| e^{j\theta} e^{j2kz} = |\Gamma_L| e^{j(\theta+2kz)}$$

The maximum and minimum values of $1 + \Gamma(z)$ occur when $e^{j(\theta+2kz)} = \pm 1$ so

$$|V|_{\max} = |V_+| (1 + |\Gamma_L|)$$

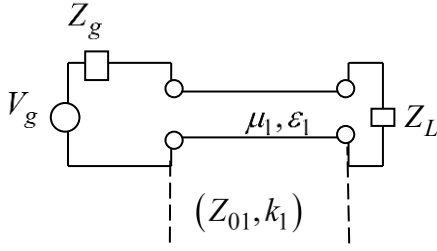
$$|V|_{\min} = |V_+| (1 - |\Gamma_L|)$$

$$\therefore VSWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} \quad \text{or} \quad = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad \text{since} \quad |\Gamma| = |\Gamma_L|$$

$$\text{where } \Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Transmission Line Circuit

Plane Wave Dual Structure



Total Voltage and Current

$$V(z) = V_+ e^{-jkz} (1 + \Gamma(z))$$

$$I(z) = \frac{V_+}{Z_0} e^{-jkz} (1 - \Gamma(z))$$

with

$$\Gamma(z) = \frac{V_-}{V_+} e^{j2kz} = \Gamma_L e^{j2kz}$$

Line impedance

$$Z(z) = \frac{V(z)}{I(z)} = Z_0 \frac{1 + \Gamma(z)}{1 - \Gamma(z)}$$

so

$$\Gamma(z) = \frac{Z(z) - Z_0}{Z(z) + Z_0}$$

$$Z(-l) = Z_0 \frac{Z_L + jZ_0 \tan(kl)}{Z_0 + jZ_L \tan(kl)}$$

 or

Reflection of reflection coeff.

$$\Gamma(z) = \Gamma(z') e^{j2k(z-z')}$$

Continuity

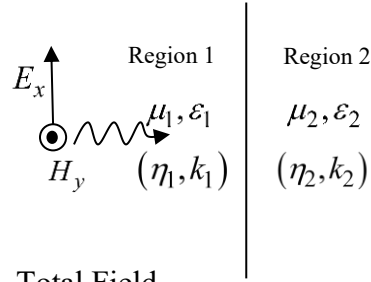
$$Z(z^-) = Z(z^+)$$

$$I(z^+) = I(z^-)$$

$$V(z^+) = V(z^-)$$

Voltage Standing Wave Ratio

$$VSWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$



Total Field

$$E_x = E_+ e^{-jkz} (1 + \Gamma(z))$$

$$H_y = \frac{E_+}{\eta} e^{-jkz} (1 - \Gamma(z))$$

with

$$\Gamma(z) = \frac{E_-}{E_+} e^{j2kz} = \Gamma_L e^{j2kz}$$

Total Field impedance

$$Z(z) = \frac{E_x}{H_y} = \eta \left(\frac{1 + \Gamma(z)}{1 - \Gamma(z)} \right)$$

so

$$\Gamma(z) = \frac{Z(z) - \eta}{Z(z) + \eta}$$

or

$$Z(-l) = \eta \frac{Z_L + j\eta \tan(kl)}{\eta + jZ_L \tan(kl)}$$

Reflection of reflection coeff.

$$\Gamma(z) = \Gamma(z') e^{j2k(z-z')}$$

Continuity

$$Z(z^-) = Z(z^+)$$

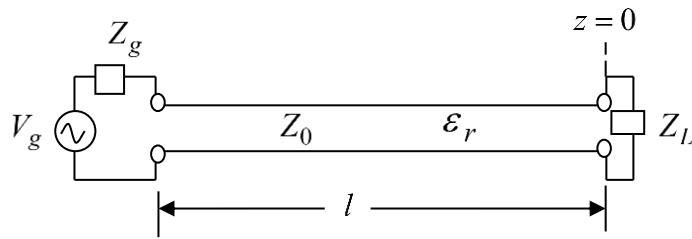
$$H_t(z^+) = H_t(z^-)$$

$$E_t(z^+) = E_t(z^-)$$

Electric Field Standing Wave Ratio

$$ESWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

Example/



given

$$V_g = 100e^{j0^\circ} \text{ volts (RMS)}$$

$$\text{freq} = 20 \text{ MHz}$$

$$\epsilon_r = 2.0$$

$$l = 6.33 \text{ meters}$$

$$Z_L = (36 + j20)\Omega = 41.2e^{j29.05^\circ}\Omega$$

$$Z_g = 50\Omega$$

$$Z_0 = 50\Omega$$

(a) Find l in wavelengths – is this a transmission line?

(b) input impedance of the line ($Z(-l)$)

(c) power delivered to the line

(d) power delivered to the load

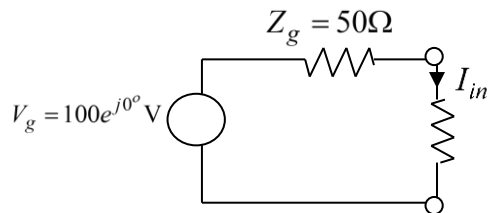
$$(a) \quad k = \frac{2\pi}{\lambda} = \frac{\omega}{v} \Rightarrow \lambda = \frac{v}{f} = \frac{c}{\sqrt{\epsilon_r} f} = \frac{3 \times 10^8}{\sqrt{2} \cdot 20 \times 10^6} = 10.55 \text{ M}$$

$$\boxed{\frac{l}{\lambda} = \frac{6.33 \text{ M}}{10.55 \text{ M}} = 0.6 \Rightarrow l = 0.6\lambda} \quad (\text{rule of thumb}) > 0.01\lambda \Rightarrow \text{yes it is a transmission line}$$

$$(b) \quad Z(-l) = Z_0 \frac{Z_L + jZ_0 \tan(kl)}{Z_0 + jZ_L \tan(kl)} ; kl = \frac{2\pi}{\lambda}(0.6\lambda) = 1.2\pi = 216^\circ$$

$$= (50) \left[\frac{(36 + j20) + j(50) \tan(1.2\pi)}{50 + j(36 + j20) \tan(1.2\pi)} \right] = (70.5 + j28)\Omega$$

$$= 75.8e^{j21^\circ}\Omega$$



$$(c) \quad I_{in} = \frac{V_g}{Z_g + Z_{in}} = \frac{100}{50 + 70.5 + j28} = \underline{\underline{0.808e^{-j13.1^\circ} \text{ A}}}$$

$$\begin{aligned} S_{av,in} &= \frac{1}{2} \text{Re} \{ V_{in} I_{in}^* \} \\ &= \frac{1}{2} \text{Re} \{ Z_{in} I_{in} I_{in}^* \} \\ &= \text{Re} \{ 37.9 e^{j21^\circ} (0.808)^2 \} \end{aligned}$$

$$\boxed{S_{av,in} = 23.1 \text{ W}}$$

(d) Recall

$$I(z) = \frac{V^+}{Z_0} e^{-jkz} (1 - \Gamma(z)) = I^+ e^{-jkz} (1 - \Gamma(z))$$

at $z = -l$

$$I(-l) = I_{in} = I^+ e^{-jk(-l)} (1 - \Gamma(-l))$$

$$I^+ = \frac{I_{in} e^{+jk(-l)}}{1 - \Gamma(-l)}$$

$$\Gamma(-l) = \frac{Z(-l) - Z_0}{Z(-l) + Z_0} = \frac{70.5 + j28 - 50}{70.5 + j28 + 50}$$

$$\Gamma(-l) = 0.212 + j0.177$$

$$I^+ = \frac{0.808 e^{-j13.1^\circ} e^{-j216^\circ}}{1 - (0.212 + j0.177)} = \underline{\underline{1e^{-j215.4^\circ} \text{ A}}}$$

$$I_L = I^+ (1 - \Gamma(0))$$

$$\Gamma(0) = \frac{Z(0) - Z_0}{Z(0) + Z_0} = \frac{36 + j20 - 50}{36 + j20 + 50} = \underline{\underline{0.28e^{j111.9^\circ}}} = \Gamma(0)$$

$$\underline{\underline{I_L = 1.133e^{-j228.5^\circ} \text{ A}}}$$

$$S_{av,L} = \frac{1}{2} \text{Re} \{ I_L I_L^* Z_L \} = \frac{1}{2} \text{Re} \left\{ \left(1.133e^{-j228.5^\circ} \right) \left(1.13e^{j228.5^\circ} \right) \left(41.2e^{j29.05^\circ} \right) \right\}$$

$$\boxed{S_{av,L} = 23.1 \text{ W}}$$